Liveness

“You know you're in love when you can't fall asleep because reality is finally better than your dreams.”
– Dr. Seuss



1. Fundamental Property

- There are 2 fundamental properties in distributed systems:
  - **Liveness**: will guarantee that something good will eventually happen
  - **Safety**: something bad will never happen
- Safety:
  - At least one consumer for a partition
  - Guarantee the order of message in a partition
  - A lot of missing configurations with default values
    but Kafka still runs correctly



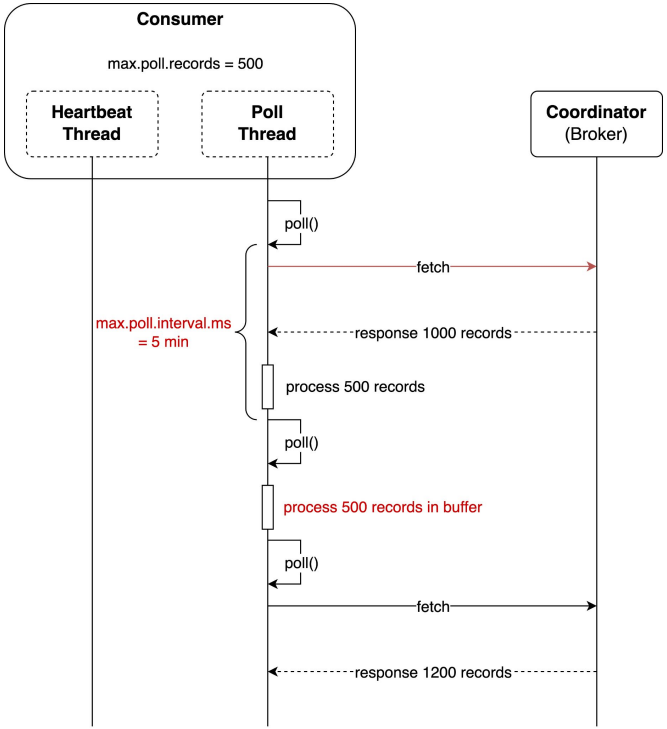
2. Liveness

- Kafka satisfies the liveness in 2 ways:
  - **Checking availability**:
    - Is it alive still?
    - Requiring consumers to periodically exchange heartbeats with the group coordinator.
  - **Checking progress**:
    - Is it still working?
    - **Being alive does not mean it is still working.**
    - Requiring consumers to invoke poll() periodically.
- There are 2 internal threads in a consumer
  - Heartbeat Thread
  - Poll Thread



3. Poll Thread

- Poll thread and heartbeat thread work independently
- By default, if a consumer does not invoke poll()
  in **5 minutes (max.poll.interval.ms)**
  → the consumer leaves the group.



4. Heartbeat Thread

- By default, consumers send a heartbeat **every 3 seconds (heartbeat.interval.ms)**
- If the coordinator does not receive a heartbeat within **45 seconds (session.timeout.ms)**, it will assume that the consumer died.
- Recommended: session.timeout.ms > 3 x heartbeat.internal.ms
- If heartbeat.internal.ms and session.timeout.ms are configured shorter, it might overwhelm brokers and cause false-positive failure

